

Scalability & Future Plan

Date	3 July 2026
Team ID	SWTID-2026-9706
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No persistent storage (session state only, by design)	Results are lost once the browser/session closes; no history across visits	Medium
2	Local-only execution (no cloud deployment)	Every user must install Python, ffmpeg, and dependencies themselves	High
3	Single, fixed reference concept bank (3 concepts)	Limited to Machine Learning, Cloud Computing, and Neural Network topics only	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single user, single local instance at a time	Deploy on a cloud host with adequate RAM (Whisper/PyTorch are memory-intensive) to support multiple concurrent sessions
2	Data Storage	In-memory Streamlit session state only, no persistence	Introduce a lightweight database (SQLite/PostgreSQL) to store historical evaluations per user
3	Performance	First analysis in a session is slower (one-time model load); subsequent analyses are fast (~0.5s for audio features)	Use pre-warmed containers in deployment, or a smaller/quantized Whisper model to reduce cold-start latency
4	Security	No authentication; purely local processing	Add user authentication and HTTPS if ever deployed publicly; validate/sanitize uploaded file types and sizes

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Expand reference concept bank (multi-concept evaluation)	1-2 months	Broader subject coverage beyond the current 3 concepts
Phase 2	Persistent storage & progress tracking	2-3 months	Track a student's improvement across multiple sessions over time
Phase 3	Multilingual support	2-3 months	Accessibility for non-English speakers
Phase 4	Real-time voice input & adaptive feedback	1-2 months	Immediate feedback during live practice, with personalized guidance based on performance patterns